

HW04 Main Report - AI Automation Testing

1. Student and Scope

- **Student:** Nguyễn Tấn Thắng - 23127259
- **Selected features:** FR-02 Login and Account Lockout, FR-07 Shopping Cart, FR-16 Product Import from CSV
- **Tooling:** Playwright, Node.js, Playwright HTML/JSON reporter
- **Browsers:** Chromium, Firefox, WebKit
- **SUT services:** Backend :3000 , Web :5173 , Admin :5174

The three features are the required one-per-pool selection from Pools A, B, and C. The final suite contains 45 unique cases: FR-02 has 16, FR-07 has 17, and FR-16 has 12.

2. Requirements Traceability

Feature	Primary SRS coverage	Test evidence
FR-02	Valid/invalid login, HTML5 email, increment by one, threshold three, 30-second lock, reset, secure password handling, FR-21/22 form rules	tests/fr02-login.spec.js TC01-TC16
FR-07	Empty state, navigation, product/quantity/subtotal/total, removal, checkout, duplicate add, confirmation, +/- controls	tests/fr07-cart.spec.js TC01-TC17
FR-16	Valid import, preview, template, extension, row validation, admin authorization, rollback, positive price, RFC 4180	tests/fr16-import-csv.spec.js TC01-TC12

3. Automation Architecture

Page Object Model

LoginPage , CartPage , and AdminImportPage own locators and user flows. A significant correction was made to CartPage : after arranging products, the suite clicks the React Router cart link instead of calling page.goto('/cart') . A full reload destroys the in-memory CartContext and previously produced false failures.

The admin POM was also corrected to use the actual login placeholders and Login button, then open the Products tab before locating the import panel. All import locators are scoped to that panel so the preview table is not confused with the product table.

Data-Driven Testing

Credentials, product rows, messages, boundaries, API payloads, and expected labels live in `tests/data/*.csv` or `*.json`. The specs iterate CSV rows and reference JSON payloads; no test-only fixture is imported without use. FR-16 uploads three physical CSV fixtures, including an RFC 4180 quoted-comma case.

Assertion Patterns

The suite uses more than three patterns: URL/navigation, visibility/text, count, HTML attribute and validity, API status/body, exact money calculation, modal event, and backend state through the admin API.

4. Human Review and Corrections to AI Output

The initial AI-generated suite contained several false-positive/false-negative risks:

1. **Unused DDT:** CSV files were loaded but their rows did not drive tests. The final suite loops through external rows and moves inline API payloads to JSON.
2. **Lost cart state:** `page.goto('/cart')` reset React state. It was replaced by SPA navigation through the header.
3. **Broken admin selectors:** The AI assumed `type="email"` and Vietnamese button text on the admin login. Selectors now match stable placeholders and the actual route flow.
4. **Unscoped table selector:** `table.bg-white tbody tr` matched both preview and product tables. The locator is now scoped to the CSV import card.
5. **Weak assertion:** Checking only for `d` did not verify totals. The final suite parses and compares exact amounts from CSV.
6. **Missed logout bug:** The old test never asserted the second/third response, so early logout passed. The final test asserts the full status sequence and reads counter/lock time through the admin API.
7. **Environment failures mislabeled as bugs:** Firefox/WebKit executables were installed and the suite rerun. Browser-launch failures are no longer present.

5. Multi-Browser Results

Each feature ran independently on Chromium, Firefox, and WebKit. Every report contains `Run by: 23127259` and an ISO timestamp; nine screenshots in `evidence/report-screenshots/` verify visible attribution.

Feature	Chromium	Firefox	WebKit	Unexpected failures
FR-02	6 Pass / 10 @bug Fail	6 / 10	6 / 10	0
FR-07	11 Pass / 6 @bug Fail	11 / 6	11 / 6	0
FR-16	7 Pass / 5 @bug Fail	7 / 5	7 / 5	0

The matrix totals 135 executions: 72 passed and 63 intentional failures corresponding to 21 failing cases across three browsers. Results are consistent across engines.

6. Defect Findings

Twenty unique SUT defects were confirmed. Examples include early/overlong account logout, plaintext password exposure, duplicate cart rows, missing removal confirmation, non-atomic CSV import, non-admin import access, and incorrect RFC 4180 parsing. Details, exact expected/actual results, severity, test mapping, and evidence paths are in `bug-report/BUG_REPORT.md`.

7. Reproducibility

```
npm run test:multibrowser:all
node scripts/verify-report-banner.mjs
```

`scripts/run-multibrowser.mjs` classifies red @bug cases separately from unexpected failures and returns non-zero only for infrastructure or non-bug failures. This prevents known product defects from stopping the matrix while still detecting a broken suite.

8. Remaining Manual Work

No test case remains unautomated. The remaining submission work requires student identity and external actions: record/upload the two videos, create GitHub Issues with the generated screenshots, and open/merge a real Pull Request. These actions are deliberately not claimed as completed in this report.